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Original Article

Low dose iodine-125 brachytherapy and transpupillary thermal therapy for small- to medium-sized choroidal melanoma

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ABSTRACT

PURPOSE: COMS established iodine-125 (^{125}I) brachytherapy as standard of care for medium-sized choroidal melanomas. However, there remains a gap in the literature regarding reduced dosing for medium-sized choroidal melanomas and standard dosing for small-sized choroidal melanomas. We explored the impact of reduced dose ^{125}I brachytherapy on local tumor control and ocular adverse events (AEs).

METHODS: Patients diagnosed 2005–2019 with choroidal melanoma <5 mm thick and managed with ^{125}I brachytherapy dose of 65 Gy to depth of 5 mm were included. Dose metrics were calculated using Monte Carlo-based dosimetry. Kaplan-Meier method was used to estimate overall survival and radiation-related AEs. Distant metastasis and local recurrence were reported using cumulative incidence with death as a competing risk. Univariate and multivariate associations were analyzed using Cox proportional hazards regression.

RESULTS: 143 patients met inclusion criteria with median follow-up of 7 years. Five-year Kaplan-Meier estimates of ocular AEs were 59.6% radiation maculopathy, 42.0% cystoid macular edema, 55.2% nonproliferative radiation retinopathy, 10.8% proliferative radiation retinopathy, and 35.6% radiation papillopathy. Five-year cumulative incidence of local recurrence was 0%, 5-year cumulative incidence of distant metastasis was 6.1%, and 5-year Kaplan-Meier estimate of overall survival was 96.5%.

CONCLUSION: Reduced dose ^{125}I brachytherapy is feasible with excellent local tumor control.

Keywords: Ocular melanoma; Choroidal melanoma; Iodine-125 brachytherapy; Reduced dose brachytherapy; Transpupillary thermal therapy; Tumor control; Ocular adverse events; COMS plaques; Monte-Carlo dosimetry

Introduction

Uveal melanoma is the most frequent primary malignant intraocular tumor in adults, accounting for 3–5% of all melanomas (1). The choroid is the most frequent anatomic site of involvement, and the COMS trial established brachytherapy as a globe-preserving standard of care treatment with equivalent patient survival to enucleation for medium-sized tumors (2,3). Clinicians have further expanded the use of brachytherapy for treatment of small and some large uveal melanomas (4). The most frequently used isotopes are iodine-125 (^{125}I) and ruthenium-106 (^{106}Ru) due to availability and favorable dose distribution (5). Of

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